



ST JOHN'S
College

Starting *Small* with AI: Building Trustworthy KPI Dashboards

Elmer Yglesias · St. John's College · Elmer.Yglesias@sjc.edu · elmerdata.ai · ORCID: 0009-0004-9538-2159



A free, open-source Python pipeline that validates any KPI against 10 rigorous steps — governed, explainable, and built for lean IR offices that need trustworthy dashboards without vendor dependency.⁴

Not a BI tool — a governance-first AI validation layer for institutional KPI pipelines that sits upstream of any dashboard platform.

kpi_parser.py

Plain language intake & structured field extraction

[CLAUDE API · Requires Claude API key]

— FEATURES

- Paste or type KPI in any format
- Three historical data options
- Claude extracts all structured fields
- Human review before validation runs
- Saves to parsed_kpi.json → hands off

PRODUCES

→ parsed_kpi.json **structured fields for engine input**

kpi_readiness_10step.py

10-step validation engine — the load-bearing core

[PURE PYTHON · NO API KEY REQUIRED]

— I. DEFINE

- 01 Register Metadata
- 02 Validate Definition
- 03 Entity Coherence

Describe the KPI precisely

— II. VALIDATE

- 04 Data Quality Screening
- 05 AI Use Assessment
- 06 Governance Check
- 07 Readiness Decision
- 08 Summary Report

Test for quality and governance

— III. ENHANCE

- 09 SMART + Tufte¹²
- 10 Industry Best Practices³

Advisory: design quality and standards

PRODUCES

→ _report.json **complete validation report**
→ _summary.csv **executive summary per step**

kpi_improver.py

Refinements, dashboard & NLP survey analysis

[CLAUDE API · Requires Claude API key]

— FEATURES

- AI-suggested definition refinements
- Board-ready HTML dashboard
- Chart.js trend visualization
- LLM survey comment analysis
- Synthetic data if none available

PRODUCES

→ _dashboard.html **board-ready HTML dashboard**
→ _suggestions.txt **AI refinement candidates**

An example of a pipeline run on First-Year Retention Rate from plain-language input to board-ready dashboard output.

1 Getting Started

Any office can start today

OPTION 1
Include data
in description

OPTION 2
Upload Excel
or CSV file

OPTION 3
No data yet —
synthetic generated

Step 1 Paste plain-language KPI description — any format

Step 2 Choose historical data option above

Step 3 Claude API extracts all structured KPIInput fields

Step 4 Human review required — confirm before validation

Step 5 Saves parsed_kpi.json → hands off to engine

Governance Proof Moment

AI never runs unsupervised. Every parsed field is displayed for human confirmation before validation begins. Step 6 is a hard gate — incomplete governance blocks publication.

Installation

LEVEL 1
Core only
Python 3.8+
No dependencies

LEVEL 2
With parser
pip install
anthropic
+ API key

LEVEL 3
Full pipeline
+ openpyxl
+ matplotlib

github.com/ElmYgl/kpi_readiness_10step

MIT License · Python 3.8+ · no external dependencies for core validation.
Scan QR for code: README & supplementary materials

2 Demo Example

Validation Run: end to end

First-Year Retention Rate

Returning FTFT Students ÷ FTFT Cohort × 100

Step 01 **PASS** All metadata fields present

Step 02 **CLEAR** Definition precise, ÷ operator confirmed

Step 03 **LOW RISK** Single source IPEDS, no join required

Step 04 **PASS** No anomalies — $z < 2.5\sigma$ across 6-year series

Step 05 **GOVERNED** NLP permitted for survey comments

Step 06 **COMPLETE** Owner, steward, equity risks documented

Step 07 **READY ✓** Ready for Dashboard Use

Step 08 **COMPLETE** Structured JSON report generated

Step 09.1 **5/5 SMART** All SMART criteria met

Step 09.2 **4/5 TUFTE** Tufte 80% — add comparison target

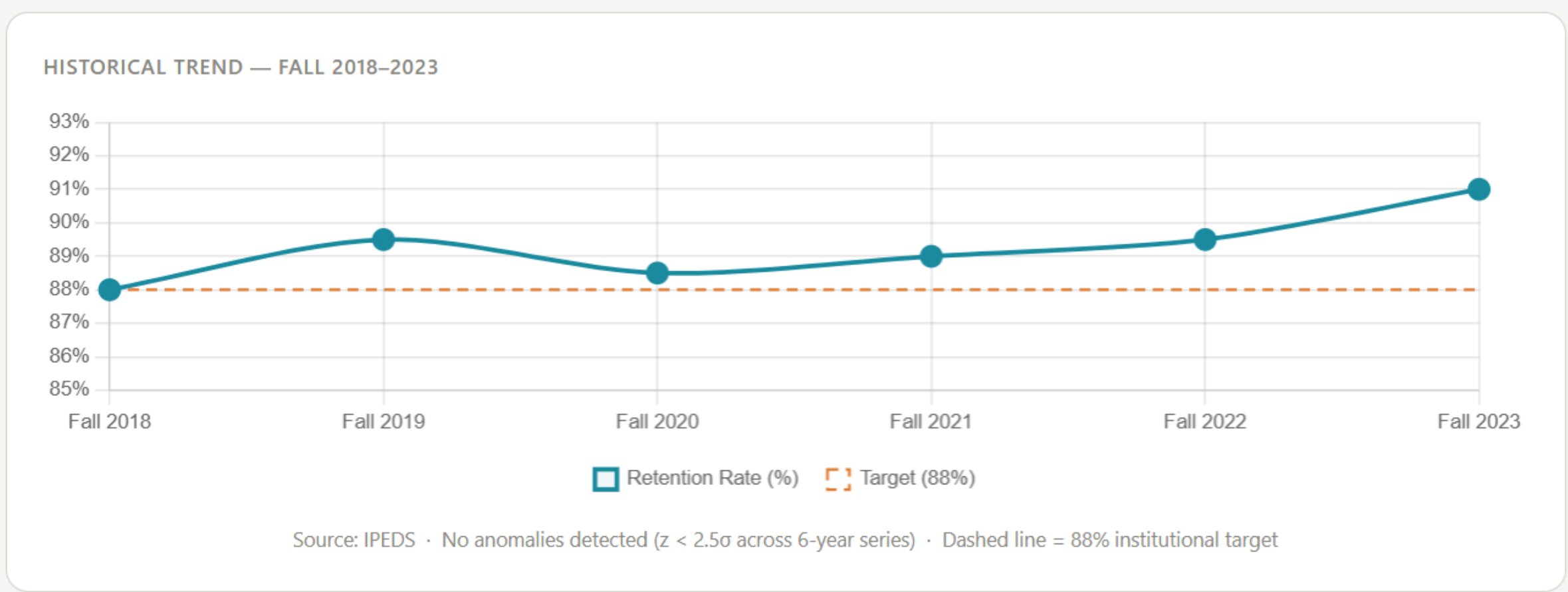
Step 10 **8/9** Strong alignment — EDUCAUSE / AGB / NACUBO

5/5
SMART

4/5
TUFTE 80%

8/9
STANDARDS

3 Output & Next Steps



Governance Architecture: oversight at every point

EDUCAUSE ³	AGB ³	NACUBO ³
✓ Data ownership & stewardship	✓ Board-level audience defined	✓ IPEDS-aligned definitions
✓ Equity risks documented	✓ 3+ year trend data	✓ Multi-year historical data
✓ Known limitations stated	✓ Target or peer benchmark	✓ Formula + source auditable

✓ PERMITTED ⁴	✗ EXCLUDED
Entity resolution	Automated individual decisions
Anomaly detection	Black-box scoring
NLP survey comments	Auto-correction without review
Synthetic data (labeled)	Ungoverned AI output

Learning Outcomes

- LO 1 Identify which AI techniques are appropriate for KPI validation — and which must be excluded — with documented rationale
- LO 2 Apply the three-entry-level pipeline to begin validating KPIs within their own office context
- LO 3 Implement a reusable 10-step KPI validation framework aligned with EDUCAUSE, AGB, and NACUBO standards

“Validation that once took hours, days, or weeks now runs in seconds.”

Piloted at Norco College · Ashby, H. & Yglesias, E. (2026). "From Strategy to Screens: Designing Dashboards that Support Decision-making," REACH Research Analytics Summit, Newport, RI.

References

- ¹ Doran, G. T. (1981). There's a S.M.A.R.T. way to write management's goals and objectives. Management Review, 70(11), 35–36.
- ² Tufte, E. R. (2001). The Visual Display of Quantitative Information (2nd ed.). Graphics Press. · Few, S. (2006). Information Dashboard Design. O'Reilly.
- ³ EDUCAUSE Data Governance Framework (2023) · AGB Dashboard Metrics for Boards (2022) · NACUBO KPI Framework (2024)
- ⁴ NIST AI Risk Management Framework (2023). NIST AI 100-1. National Institute of Standards and Technology.